

Formula and Audit Sheet

Permitted reference sheet for course practice and written examinations

Prepared for Yasin Kaymaz, PhD | Version 1.0 | 14 September 2026

This is a teaching package, not a clinical, legal or institutional accreditation document. Scientific sources are identified by [Sxx] in the Course Source Register.

Basic calculations

Quantity	Formula / interpretation
Known-base GC	Count(G or C) / count(A,C,G,T). Undefined when there are no known bases under this course contract.
BED interval length	end - start; start included, end excluded.
One-based inclusive length	end - start + 1. For a positive-length BED interval, inclusive start = BED start + 1; end unchanged.
Nominal average depth	Total sequenced bases / target bases. N paired fragments of read length L per end yield N x2xL sequenced bases.
Usable mean depth	Nominal mean depth x explicit usable fraction (simplified teaching model).
Fold change	Ratio B/A, with a stated normalization and zero-handling rule. $\log_2(2)=1; \log_2(1)=0; \log_2(0.5)=-1$.
Toy TPM	$\text{rate}_i = \text{count}_i / \text{length}_i$ in kb; $\text{TPM}_i = \text{rate}_i / \text{sum}(\text{rates}) \times 1,000,000$. This does not create count-model input.
Sensitivity / recall	$\text{TP}/(\text{TP}+\text{FN})$: among actual positives, how many were detected?
Specificity	$\text{TN}/(\text{TN}+\text{FP})$: among actual negatives, how many were correctly called negative?
Accuracy	$(\text{TP}+\text{TN})/(\text{TP}+\text{TN}+\text{FP}+\text{FN})$. Inspect class balance and independent sample count.
Balanced accuracy	$(\text{Sensitivity} + \text{specificity})/2$ for a binary task. Undefined denominators need a stated rule.

BH adjustment

Sort m p-values ascending. At ordered rank i, calculate $m \times p(i)/i$. Starting at the largest rank, take a cumulative minimum moving backward and cap at 1. Return values to the original order. Reject those with adjusted value $\leq$ the selected FDR level. State the family of tests. The original guarantee is for independent tests; broader settings need justified assumptions. [S08]

Worked values for four tests: raw 0.001, 0.010, 0.040, 0.200 → adjusted 0.004, 0.020, 0.05333, 0.200. At 0.05, the first two pass. A p-value is not $P(\text{null} | \text{data})$; FDR is not the posterior probability that a particular gene is false. [S07, S08]

Interpretation boundaries

A result	Does not by itself establish
High RNA count	High protein abundance or activity.

A result	Does not by itself establish
High average depth	Every base passes a coverage requirement.
High alignment identity	Shared function or a probability of homology.
A compact cluster	A validated cell type or disease mechanism.
High test accuracy	Generalization if donors or test information leaked.
High pLDDT	Affinity, variant pathogenicity or experimental validation.
ClinVar review stars	A numerical probability of pathogenicity.
Names removed	Anonymous data or permission for every destination.

GENOMES

Letter	Check	Record
G	Genome / reference / annotation	Accession and version, or justified N/A for non-sequence work.
E	Environment / execution	Python/runtime version, seed, analysis steps and clean rerun.
N	Numbers / dimensions	Rows, columns, samples, features and expected sizes.
O	Offsets / coordinates	0- or 1-based; inclusive or half-open; units and conversions.
M	Multiplicity / experimental unit	What is independent? What family of tests was considered?
E	Entities / identifiers	Unique IDs, sample matching, transcript versions, no silent row-order assumptions.
S	Sources / scope	Provenance, real versus synthetic, citations, permitted use and limitations.